

Zadanie 3 - Robert Mesek

spline2D_comp.m

```
function spline2D_comp()
% [...]
function spline2Duniform()
knot_vectorx = [0 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15;]
knot_vectory = [0 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15;]

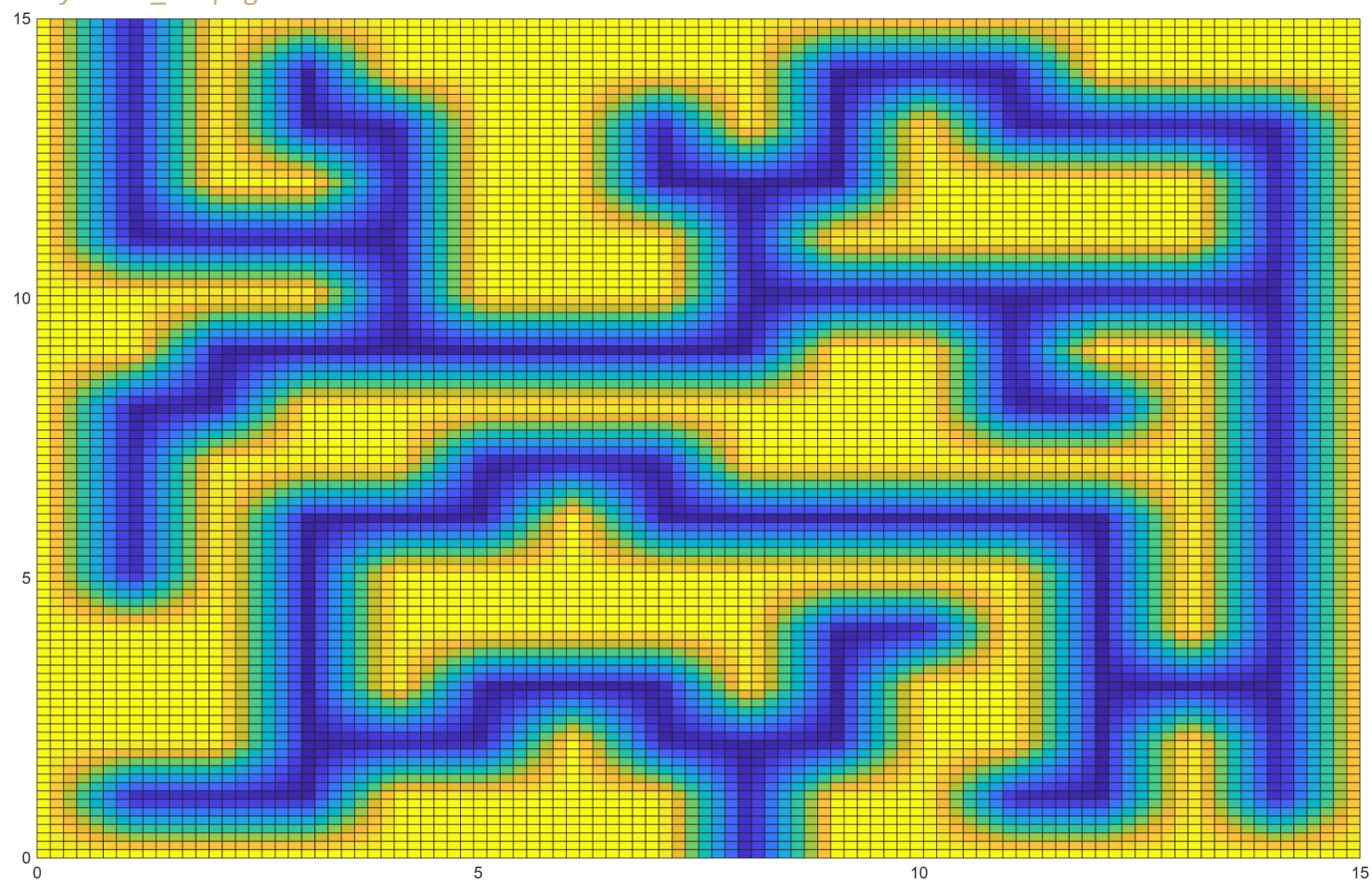
coeffs = [
    1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1;
    1 0 1 1 1 0 0 0 0 1 1 0 0 0 0 0 0;
    1 0 1 1 1 1 1 1 0 0 1 0 1 1 1 1 1;
    1 0 0 0 0 0 0 1 1 0 1 0 1 0 0 1;
    1 1 0 1 1 1 0 1 1 0 0 0 0 0 1 1;
    1 1 0 0 1 1 0 0 1 0 1 1 1 1 1 1;
    1 1 1 0 1 1 1 0 1 0 1 1 1 1 1 1;
    1 1 0 0 1 1 0 0 1 0 1 1 0 0 1 1;
    0 0 0 1 1 1 0 1 1 0 0 0 0 1 1 1;
    1 1 0 0 0 1 0 1 1 1 0 1 0 0 0 1;
    1 1 1 1 0 1 0 1 1 1 0 1 1 1 0 1;
    1 0 1 1 1 1 0 1 0 0 0 1 1 0 0 1;
    1 0 0 0 0 0 0 1 0 1 0 1 1 0 1 1;
    1 1 1 0 1 1 1 1 1 1 0 1 1 0 1 1;
    1 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1;
    1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1;
]

precision = 0.01
% [...]
%X and Y coordinates of points over the 2D mesh
[X,Y]=meshgrid(x,y);
Z = zeros(size(X));

hold on
for i=1:nrx
    %compute values of
    vx=compute_spline(knot_vectorx,px,i,X);
    for j=1:nry
        vy=compute_spline(knot_vectorx,py,j,Y);
        Z = Z + coeffs(i,j) * vx .* vy;
    end
end
surf(X,Y,Z);
hold off

end
% [...]
```

labyrinth_2d.png



labyrinth_3d.png

